

Project Portfolio

Jaime Uy De Baron

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1 Custom Hip Implant Approachability Tools

Zimmer Biomet, a Fortune 500 medical device company, tasked me with developing tools to improve the approachability of implants within their custom hip division, Ossis. Twenty-one completed cases reported difficult-to-install implants, impacting surgical efficiency and patient outcomes. Improving implant approachability was therefore critical to bettering both surgical performance and patient care.

Key Responsibilities

- Analyzed 500 historical surgical cases to identify trends influencing screw approachability
- Developed quantitative approachability models tailored to each surgical approach
- Validated models against active, work-in-progress implant designs
- Produced comprehensive documentation outlining methodology, tool usage, and integration of future datapoints
- Conducted 3D scanning of manufactured implants to assess underbuilding and established acceptable tolerances with suppliers

Key Learnings

- Strengthened understanding of structured R&D processes within a regulated medical device environment
- Led development of a novel analytical tool in an area with no prior literature
- Recognized the critical role of rigorous documentation for scalability and long-term usability
- Gained insight into the orthopedic implant market and its key industry players
- Developed a deeper understanding of orthopedic design philosophies and surgical constraints

Outcomes

Implemented a design tool now integrated into all future custom implant workflows. Enabled real-time visualization of screw approachability during design review. Validated model predictions against previously identified approachability regions, demonstrating strong alignment with historical expert assessments.

2 RoboCup

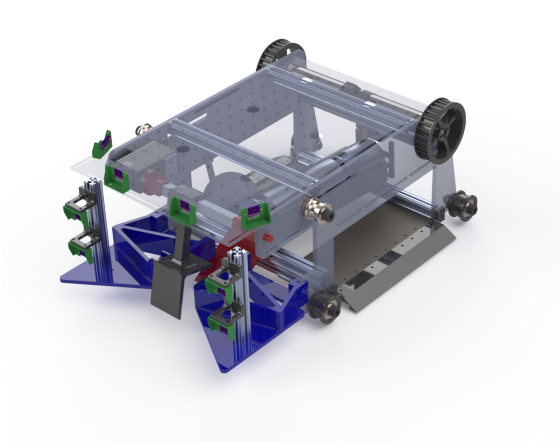


Figure 1: Render of final robot

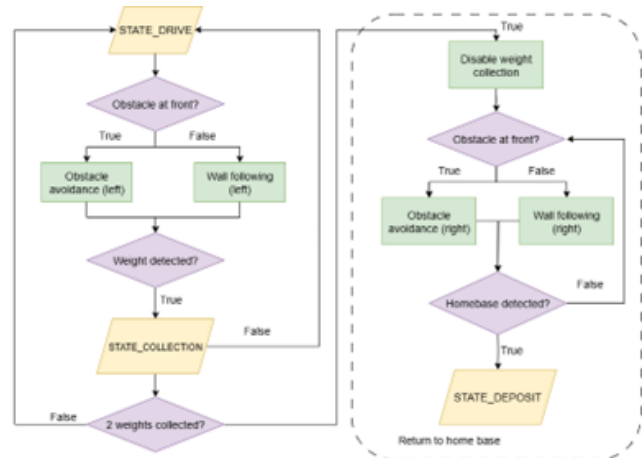


Figure 2: Snippet of navigation logic

RoboCup is a yearly robotics competition requiring fully autonomous navigation. Competitors ideate, design, build, and test robots that must navigate through obstacles, detect targets, collect them, and return them to base.

Key Responsibilities

- Led a team of three by taking on main organizational and design roles
- Manufactured custom components, including laser-cut acrylic plates and a redesigned sensor mounting system
- Developed structure of main program with 10+ modularized files
- Debugged and improved navigational logic after countless situational tests
- Integrated different mechanical movements with accurate sensor identification and software timing
- Conducted testing and iterative refinement of code and hardware to improve robot's reliability

Key Learnings

- Understood the requirements in being a project manager and team leader
- Developed proprietary prototyping philosophy for milestone-based projects
- Applied end-to-end systems engineering across mechanical, electrical, and software domains
- Strengthened rapid prototyping skills under time constraints
- Learned the importance of modular design for rapid debugging and maintenance
- Improved real-world control system tuning and sensor integration techniques
- Gained experience balancing performance, robustness, and competition constraints

Outcomes

Delivered a competition-ready robot that met all the design requirements. Robot had one of the highest rates (of all competitors) of autonomous navigation back to homebase and successful collection of target objects.

3 Histology-radiology co-location

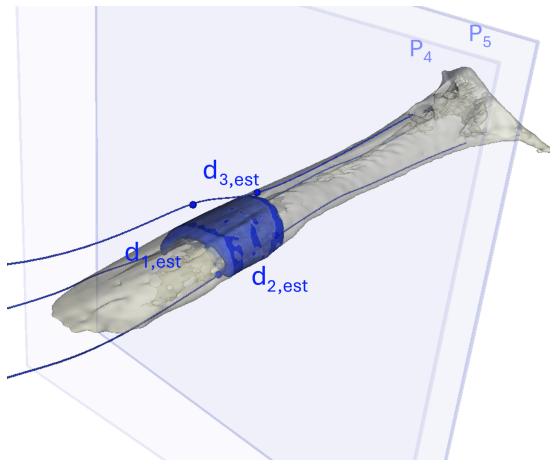


Figure 3: Calculated planes superimposed on CT bone model

The University of Otago's Future Health Research Programme involves university students taking part in PhD research. My research involved Radiology-Pathology fusion with a PhD candidate. Our research improved tumour diagnostic information by aligning post-operation histology scans with pre-operation radiology volumes.

An initial method (M1) used the cross product to construct the plane spanned by two vectors. I developed and implemented second and third (M2 and M3) methods for plane location. M2 used different vectors, while M3 introduced a third spline that locked a rotational degree of freedom and eliminated the uncertainties present in M1 and M2.

Key responsibilities

- Comprehensive literature view of previous methods and results
- Segmentation of historical and current CT scans
- Developing M2 following candidate's specification
- Ideating and programming M3
- Testing methods M2 and M3 with canine specimens

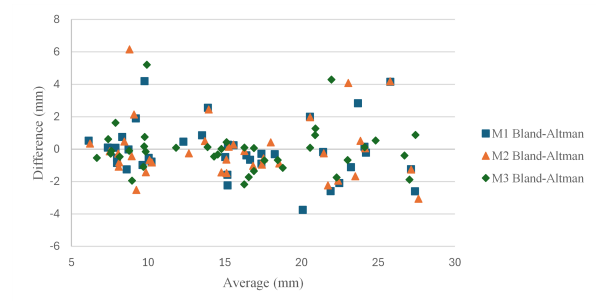


Figure 4: Bland-Altman plot of M1 vs M3 results

- Detailed documentation of work, thoughts, and development progress
- Drafting a manuscript for publication in the Journal of Physicists, Engineers, and Scientists in Medicine

Key learnings

- Medical terminology and operational procedure for tumour resection
- Specimen handling and testing procedure in a regulated environment
- Mathematical methods replicated in Python to calculate scan location
- Segmentation philosophies and methods

Outcome

During testing of canine dog bone samples, M1 produced a mean error of $0.12 (\pm 1.7)$ mm, M2 produced a mean error of $0.06 (\pm 1.9)$ mm, while M3 produced a mean error of $0.02 (\pm 1.5)$ mm. Mean errors for each method were of similar magnitude to 3D registration error reported in histology co-registration frameworks for other tissues. This study supports a simple low cost opportunity for use of digital histology images to support bone sarcoma radiology interpretation and treatment planning (PDF available on request).

4 Solar-Powered Car

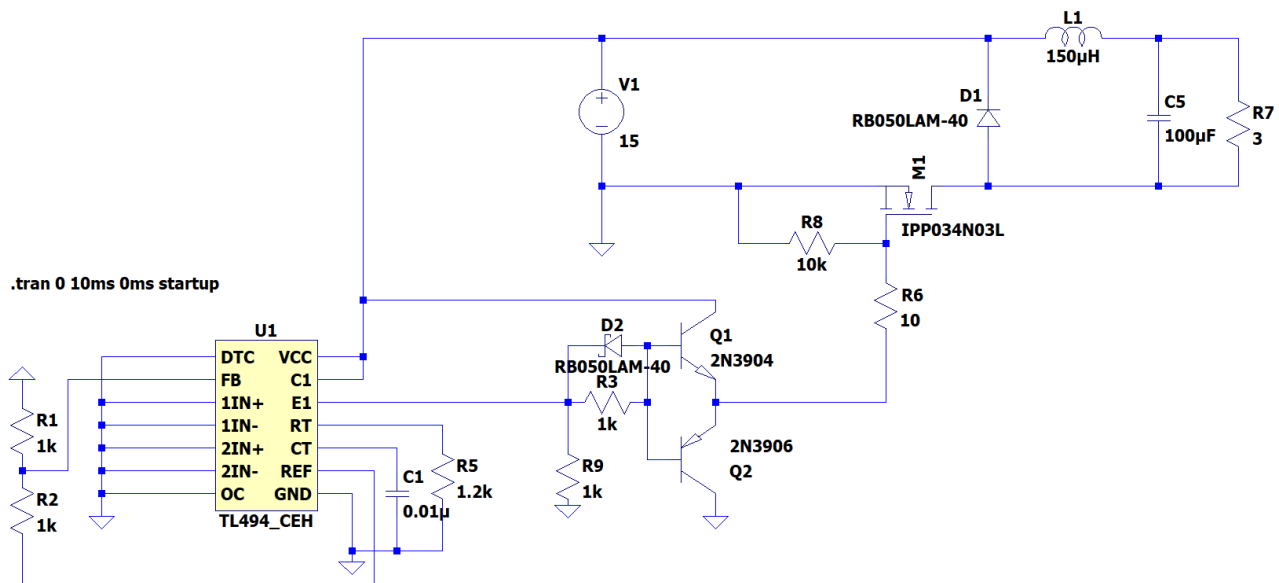


Figure 5: Buck converter schematic

Designed and implemented a high-efficiency, closed-loop buck converter to regulate solar panel output for a DC motor in a solar-powered car application. The objective was to maintain operation at the panel's maximum power point (MPP) while delivering stable voltage under variable load conditions.

Key Responsibilities

- Designed a DC-DC buck converter from first principles, including inductor sizing, air gap calculation, and capacitor selection
- Developed a closed-loop feedback control system using an integrator-based PWM controller
- Implemented a MOSFET-based gate driver to improve switching speed and reduce conduction losses
- Designed and optimized a custom PCB layout to minimize parasitic inductance, capacitance, and switching noise
- Characterized the solar panel to determine maximum power point across varying light conditions

- Validated performance through oscilloscope measurements and LTSpice simulations

Key Learnings

- Applied power electronics theory to a real-world renewable energy system
- Gained practical experience in magnetic component design (inductor windings, air gap sizing)
- Strengthened understanding of switching losses, control stability, and ripple minimisation
- Developed PCB layout strategies for high di/dt and dv/dt systems
- Learned the importance of iterative testing between simulation and hardware

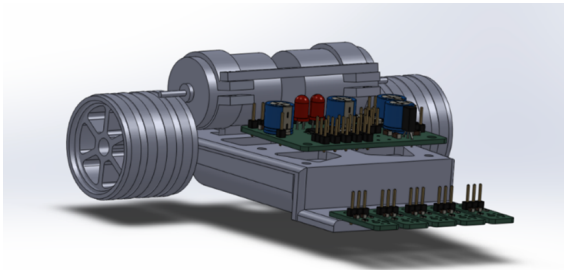
Outcomes

Delivered a functional buck converter capable of stable voltage regulation under varying solar input and load conditions. Achieved low output voltage ripple and fast switching response (34 ns switching time). Achieved the fastest switching time and highest efficiency in the entire cohort. Competed in the solar car race and won the competition with a large gap to second place.

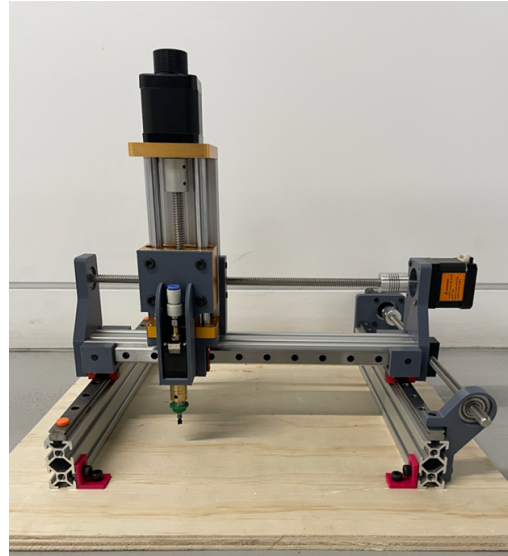
5 Line Following Robot

Designed and built a high-speed autonomous robot incorporating custom PCB design, mechanical optimization, and PID control. Optimized for speed and stability, achieving a best recorded time of 15.31s.

- Designed dual-PCB system (ATmega4808, MOSFET motor drivers, modular sensor board)
- Implemented 5-sensor PID and hybrid Bang-Bang control algorithms
- Optimized PCB layout for EMI reduction and high-current motor traces
- Iterated mechanical chassis for weight distribution and corner stability



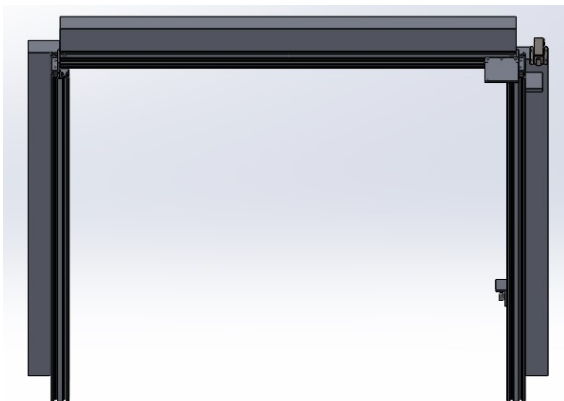
- Designed 196mm x 196mm working area within 340mm cube
- Iteratively prototyped structural components for strength and fitment
- Optimized part count to maximize accessibility and ease of assembly
- Achieved full assembly in 50 minutes



6 Takahē Recovery Gate

Designed a lightweight, remotely triggered conservation gate for DOC to safely capture endangered Takahē. Final design was equal national winner.

- Engineered aluminum rail-based drop-gate with net capture system
- Designed WiFi-enabled PCB (Raspberry Pi Pico W + solenoid actuation)
- Integrated ultrasonic safety detection
- Delivered full BOM under \$500 (budget limit \$2000)



7 CNC Pick-and-Place Machine

Developed a compact hobbyist pick-and-place machine using standard components and minimal 3D printing. Total build cost: \$208.92.

8 Elevator PLC Control System

Programmed full elevator logic using Ladder Logic and PI control.

- Implemented PI-based acceleration/deceleration control
- Designed circular-buffer scheduling algorithm
- Built FSM for door logic, movement, and prioritization

9 Embedded Systems – Step Counter

Programmed multi-mode step counter using cooperative task scheduler.

- Designed polling scheduler for multi-frequency tasks
- Implemented goal-setting and distance tracking logic
- Integrated joystick, buzzer, LED display, and ADC inputs

10 UCFK4 Game (Embedded C)

Developed two-player Rock-Paper-Scissors game on 5x7 LED matrix.

- Implemented finite state machine in C
- Optimized for responsiveness within limited MCU resources